

## Python Programming Test

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Batch: Python (2043)

Q.1> What is a string in Python?

→ A string is a sequence of characters used to represent a text.  
- string are enclosed in single quotes ('), double quotes ("), or triple quotes (").

Q.2> Which symbol is used to create a tuple?

→ '()' symbol is used to create a tuple.

Q.3> Which python data type stores unique values?

→ 'set' data type stores unique values.

Q.4> Write the syntax to create an empty dictionary?

→ `di = {}`

Q.5> Which loop is generally used to iterate over a sequence?

→ for loop

Q.6> Which keyword is used to exit a loop immediately?

→ - 'break' keyword is used to exit loop immediately.

Q.7> Can a list contain duplicate values?

→ Yes

Q.8> Can a tuple be modified after it is created?

→ No

Q.9) What is the index of first element of python list ?

→ - index of a first element is '0'

Q.10) 'len()' function

| list                                                  | tuple                                                              |
|-------------------------------------------------------|--------------------------------------------------------------------|
| - [] symbol is used to create list                    | - () symbol is used to create tuple                                |
| - list is mutable.                                    | - tuple is immutable                                               |
| - we can perform insert, remove operations with list. | - we cannot perform insert, remove oper <sup>n</sup> . with tuple. |

| set                                               | dictionary                              |
|---------------------------------------------------|-----------------------------------------|
| - stores unique values                            | - stores 'key: value' pairs             |
| - written using {}                                | - written using {key: val}              |
| - duplicate values must be automatically removed. | - key must be unique.                   |
| - syntax: {10, 20, 30}                            | - syntax: {"name": "Diksha", "age": 22} |

Q.13) → `text = "Diksha"`  
`count = 0`  
`for i in text:`  
`count += 1`  
`print ("No. of characters in {text} is {count}")`

Q.14) → `for i in range (1, 11):`  
`print (i)`

Q.15) → `i = 10`  
`while i > 0:`  
`print (i)`  
`i -= 1`

Q.16) → a) `text = "Python Programming"`  
`print (text)`

b) `print (text[0])`

c) `print (text[-1])`

d) `print (text[0:6])`

e) `print (len(text))`

Q.17) → `numbers = [10, 20, 30, 40, 50]`

a) `numbers.append (60)`

b) `numbers.remove (20)`

c) `numbers.insert (2, 35)`

d) `numbers[0:3]`

e) `len (numbers)`

Q.18) → fruits = ("Apple", "Banana", "Mango", "Orange", "Grapes")

- a) print (fruits) ✓
- b) print (fruits[2]) ✓
- c) print (fruits[0:2]) ✓
- d) print ("Apple" in fruits) ✓
- e) print (len(fruits)) ✓

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Q.19) → numbers = {10, 20, 30, 20, 40, 10, 50}

- a) duplicate values will be remove.
- b) numbers.add(60) ✓
- c) numbers.remove(30) ✓
- d) print (40 in numbers) ✓
- e) print (len(numbers)) ✓

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Q.20) → student = {"name": "Rahul", "age": 22, "course": "Python", "city": "Delhi"}

- a) print (student["name"]) ✓
- b) print (student["age"]) ✓
- c) student.update({"marks": 85}) ✓
- d) student.update({"city": "Mumbai"}) ✓
- e) student print (student.keys()) ✓

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Q.21) → A. For loop

- a) →
- for i in range (1, 11):
- print (i)

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b) → for i in range(1,21):  
if i%2 == 0:  
print(i)

B. while loop

c) →

i = 1

while i <= 10:

print(i)

i += 1

d) → i = 10

while i > 0:

print(i)

i -= 1

e) →

i = 1

total = 0

while i <= 10:

total += i

print(i)

i += 1

print(total)